Introduction

Phenobarbital (PB) is commonly used as an anticonvulsant drug due to its sedative propeties in anti-seizure management for the neonatal demographic.(1) It is typically administered as an oral elixir for paedriatics with a dosage preparation of 20mg/5ml.(2) Although its efficacy for controlling seizures has been routinely assessed, concerns regarding the generic pharmaceutical formulation has been disclosed. Primarily due to its significantly high ethanol concentration of 38% v/v (15mg/5mL), which has shown to be potentially harmful for paeditaric patients.(2) A retrospective study performed on neonates affirmed that ethanol is likely harmful due to its negative synergic effects and its possible outcome of intoxication.(3) Nonetheless, the ambiguity surrounding the chronic exposure of ethanol and its psychoactive effects on neonates showcases the importance of care in administering it and restricting its concentration in pediatric formulations.(4) Pharmaceutical excipients are crucial in enhancing the bioavailability and stability of pharmaceutical dosage forms however careful evaluation of certain therapeutic agents have to be monitored for such a vulnerable population. (5)

Alcohol degradation in neonates is significantly slower compared to adults resulting in an increased chance of endogenous ethanol accumulation in the blood plasma.(4) The consequence of a limited ethanol elimination kinetic in children poses a risk of intoxication. A study conducted by Kvigne et al assessed the impact of chronic ethanol exposure on two neonates before and after birth.(6) It was revealed that the second neonate was diagnosed with fetal alcohol syndrome as a result of a high ethanol levels of 246.5mg/dL/h and an ethanol clearance rate of 17.3mg/dL/h. In summary, the interindividual variability of ethanol elimination kinetics highlighted amongst the numerous studies available evidences why healthcare professionals have to be cautious in administering ethanol containing formulations. (7)

Appraisal

The efficacy of phenobarbital use is well established for the treatment of status epilepticus through many pragmatic clinical experiments.(8) One being a randomised placebo controlled design montherapy trial which showed the statistical and functional success of PB as the patients screened with PB showcased a significant decrease in seizures after a year. (9) Nevertheless, PB has been associated with prevalent chronic adverse effects hence its safety profile has been in question especially amongst the paediatric population.

Name of Excipient	Role of excipient
Ethanol	Solvent
Orange Oil (Benzyl Alcohol)	Flavouring
Anise Oil	Flavouring
Coriander Oil	Flavouring
Lemon Oil	Flavouring
Tartrazine	Colouring
Yellow Dye Sunset	Colouring
Glycerol	Thickener
Purified Water	Solvent

Table 2.1: The excipients present in Phenobarbital Elixir (10)

The primary endpoint of ethanol and its prominence in the formulation of PB is that it enhances the solubility of PB. In the absence of ethanol PB has poor solubility (1mg/mL) however in the presence of ethanol it is freely soluble (100mg/mL).(11) This pronounced increase in solubility is beneficial as it allows PB to be administered as an oral solution which in regard to neonates is advantageous as oral solid dosage forms are impractical and inadequate for their needs. The oral solution route is preferable as not only does it facilitate their ease of taking the medication it also provides maximal dosing flexibility as it is a lot easier to adjust the dosages of solutions than conventional rigid tablets. In spite of this , ethanol has been shown to heighten drug concentrations in the GI tract which generates a faster absorption rate resulting in variable drug plasma concentrations.(7) This is not ideal as it can lead to adverse reactions as ethanol is a cerebral depressant. (3)
Despite this, ethanol increases the stability of the elixir which in turn increases its shelf life. This is fundamental as achieving high drug stability means that the efficacy and safety of the elixir is not compromised due to any degradation impurities during its shelf life. As an unopened bottle of the elixir can last up to 36 months patients are not required to opt for regular repeat prescriptions orders frequently.(2) Despite this, a risk is posed as drug misuse can occur due to decreased monitoring of the patient.

Orange Oil terpeneless, containing benzyl alcohol (BA) , is employed in the pharmaceutical making of PB as a flavouring agent that masks the bitter taste of the elixir. This makes it more palatable for children hence increasing their chance of adherence. However, a multifold of reports displayed the unfortunate aftermaths of BA exposure to neonates from respiratory depression to fatal death.(12) As a result BA is contraindicated for the usage of neonates and children up to 3 years old. A clinical study of BA toxicity evidenced that 10 neonates were in neonatal intensive care after they experienced gasping syndrome following a marginal exposure of 130mg/kg/day. (13) This syndrome consists of respiratory distress , convulsions , metabolic acidosis and subsequent death. Upon observation of many clinical evidences and studies of BA toxicity it has been revealed that it is due to the maturational inadequacy of BA degradation. Typically, BA is rapidly converted to benzoic acid adjoined with glycine in the liver to be later excreted as hippuric acid. However , it has been noted that infants have immature livers which accounts for the numerous cases of BA toxicity.(14) As the liver is immature , there is little to no metabolic clearance of benzoic acid which leads to the accumulation load of the acid to overwhelm the functional capacity of the liver . Hence the detoxification of benzoic acid through glycine conjugation does not occur . (14) The risk of neonatal exposure inherent to BA counterbalances its positives as an excipeint. An introspective study of neonates exposed to BA and survived is needed to establish whether a dose response relationship occurs .As the elimination of BA is highly variable due to factors such as age it is complex to construct an exemption threshold. (15)

The dosage of PB is individualised according to the weight and age of the patient. This increases the likelihood of systemic error which is risky as PB is a NTI drug so doses have to be carefully titrated and monitored as there is a small window for error. (16) Patients can measure their doses wrong meaning that therapeutic outcomes wont be achieved. However, an elixir is better suited for paeditrics as tablets are impractical and they have better dosage uniformity.

Reformulation

Several premilnary studies have evaluted the efficacy of PB and its prominence in the therapeutic care of tonic clonic seizures for years since its discovery .(1) It's been highlighted as an inexpensive, effective antiepileptic drug. (17) However upon consideration of both ethanol and benzyl alcohol in section…its been noted that there are several drawbacks to the generic formulation in meeting the needs of the pediatric demographic. As a deduction , a hypothesised alcohol-free experimental formulation design will be proposed with propylene glycol replacing ethanol in a cosolvent system.(11) A cosolvent system augments the solubility between non-miscible phases and enhances the stability of the drug. Phenobarbital solubility and stability has been shown to be promoted by the presence of water miscible solvents. A literature review on an alcohol-free solution of PB detailed the acceptable cosolvents in formulations one being propylene glycol. An experiment was conducted where various quantities of cosolvents were examined to investigate the maximal solubility of PB. PB was kept constant at 400mg/mL whilst the cosolvent concentrations of USP syrup and propylene glycol were routinely adjusted . It was concluded that the formulation with propylene glycol at 10%v/v and USP syrup at 2-4%v/v had no crystal growth.This is significant as the physical instability of oral liquid formulations is associated with the formation of precipitates. Moreover, the shelf life of this alternative formulation was assessed using the Arrhenius Accelerated Stability Test which concluded that the shelf life was more than two years. (11)This observational experiment evidenced that PB can be soluble in an alcohol-free solution in the presence of propylene glycol. However the limitations of a cosolvent system is that toxological solvents that increase solubility have to be limited in paediatric formulations. For example , Propylene glycol has been proven to be successful in increasing the solubility of PB however it has been linked to neonatal toxicity in cases of chronic exposure.(18)A case study on propylene glycol , illustrated a PG assay model that was used to imitate the effects of PG when co-administered with PB. (19) This is important as it can be used as a reference for prescribers to evaluate the safety of excipients on neonates.

Conclusion

In conclusion, for the paeditaric population an empirical approach is required in preparing extrampenous formulations.(19) The revised reformulation of PB highlights why research on excipients is crucial for improving clinical safety. Although PB displays some adverse reactions in neonates its prevalent use in developing countries over the years means that it wont be completely removed from the paediatric pharmacopeia.

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